OriginX: Continuous Conditioning of a Frozen Robot Policy for RoboCasa365

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Abstract

We describe OriginX, a derivative of Xiaomi-Robotics-1-RoboCasa365 that adds continuous observation-conditioned modulation to a frozen adapted robot policy. A 4.22-million-parameter branch pools visual-language features and proprioceptive history, then changes the action transformer's conditioning during generation. It is trained with a flow objective, frozen-teacher velocity consistency, and auxiliary stage supervision; stage labels are absent at inference. A completed author-run evaluation covers all 50 RoboCasa365 target tasks with 50 episodes each: 1,496 of 2,500 episodes succeed, yielding 59.84%. This is numerically above the highest overall value displayed on the official leaderboard on October 9, 2026 (58.1%), but is an unmatched comparison, not an accepted rank. An earlier 600-pair comparison gives only a one-percentage-point difference, with a reported 95% interval of $[-2.17, 4.17]$; it does not establish a reliable improvement or isolate the branch's contribution. Secondary language-assistance studies retain 56 verified recoveries among 1,004 historical failures and no raw success-rate gain in a 50-case follow-up. A later bounded diagnostic reports X/A/B endpoint outcomes and teacher interventions on Xiaomi and GR00T, retaining initial-state deviations and incomplete GR00T coverage; it does not establish a general intervention gain. We provide the implemented adaptation recipe, complete outcome accounting, and limitations needed to reproduce and assess this author-reported result. Organizer review is pending.

1 Introduction

Adapting a generalist robot policy raises two distinct questions: how can a small learned module influence the deployed action generator, and how much of an observed evaluation score can be attributed to that module? RoboCasa365 provides a demanding setting for both questions: its multi-task benchmark evaluates atomic interactions and long-horizon composite kitchen skills, including held-out composite tasks [10–12]. A strong inherited policy makes adaptation practical, but also makes it essential to distinguish the complete system's performance from the incremental effect of a new component.

OriginX extends the public Xiaomi-Robotics-1-RoboCasa365 checkpoint [6] in two recorded stages. An Action LoRA adapter first produces an adapted policy. A second stage freezes that entire policy and learns a continuous conditioning branch from observed visual-language features and proprioception. The branch changes the action transformer's adaptive-normalization conditioning, so it remains part of action generation at test time. Auxiliary stage prediction regularizes its shared features during training; its logits do not choose a hard option or supply privileged information at deployment.

Our complete frozen-checkpoint evaluation records 1,496 successes and 1,004 failures across 2,500 planned episodes, with no missing or infrastructure-unknown outcomes. The resulting 59.84% author-run score exceeds the 58.1% highest overall value displayed on the public leaderboard at our October 9, 2026 check [18–20]. That numerical comparison motivates reporting the recipe in detail. It does not confer an official rank: OriginX is absent from the published leaderboard, its submission remains under review, and the separately reported runs are not paired.

The available negative evidence further limits attribution. An earlier comparison of the complete OriginX endpoint against the original Xiaomi policy yields 419 versus 413 successes in 600 paired trials, a one-point difference whose reported interval crosses zero. It neither validates the branch as an improvement nor separates the effects of LoRA, branch training, and the inherited checkpoint. We report that experiment alongside the complete 50-task score rather than treating a high single-arm score as a substitute for a controlled gain.

This paper contributes a concrete inference-active conditioning design with its recorded training lineage; a complete, frozen 50-task evaluation with retained failures and execution evidence; and a scoped analysis of the resulting policy's limitations. Language-assisted failure replay is a secondary investigation, followed by a smaller fresh-seed

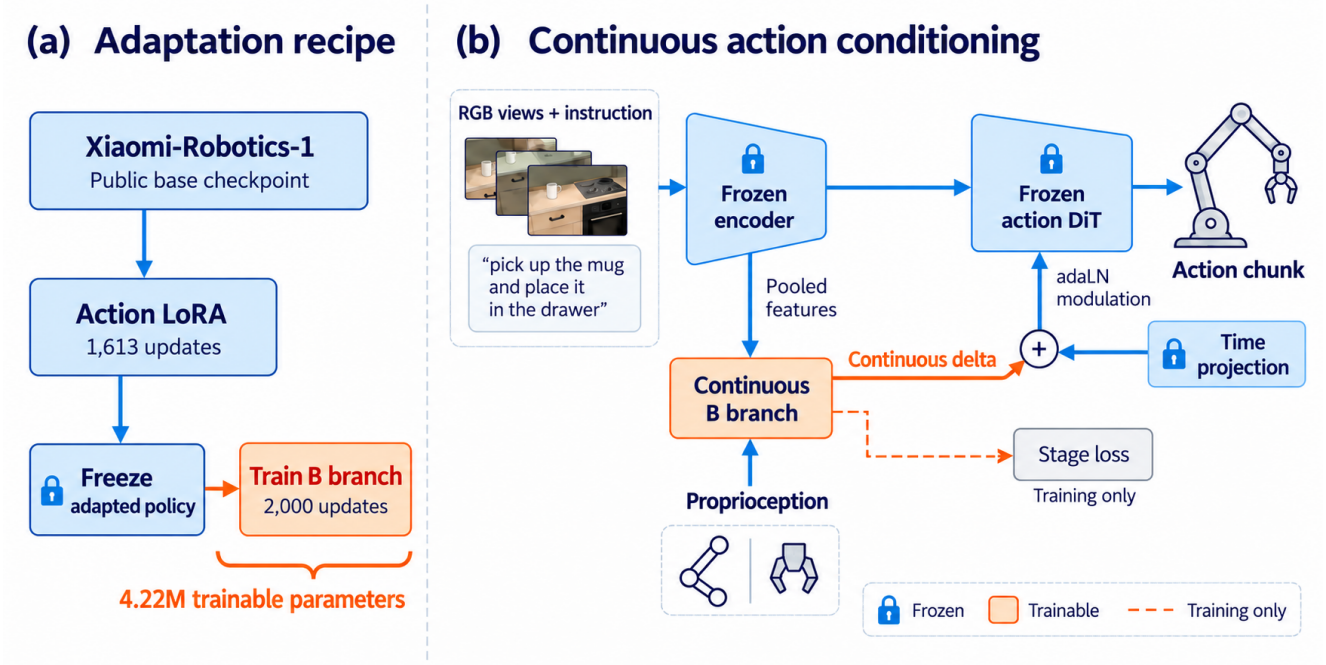


Figure 1. **Adaptation and continuous conditioning.** This AI-generated schematic was checked against the implementation; its kitchen illustrations are not evaluation frames. A2000 denotes the unchanged adapter actually trained for 1,613 updates. The 4,216,839 new B parameters include the auxiliary head. The frozen time projection takes flow time; the observation-conditioned residual changes its output before action-transformer modulation. The stage head is a parallel training-only branch, not a discrete router.

control study. Together these records make the implementation, measured performance, and remaining attribution questions inspectable.

2 Related Work

Parameter-efficient action-policy adaptation. OriginX inherits the released visual-language-action model and flow-based action generator of Xiaomi-Robotics-1 [6, 9]. Action LoRA uses low-rank weight adaptation [7]; our second stage instead freezes the adapted weights and predicts an observation-dependent residual in an existing conditioning pathway. Adaptive-normalization conditioning is established in diffusion transformers [8]. We claim neither a new flow objective nor the invention of conditional modulation. Our contribution is the specified combination in a frozen robot-policy implementation and its evaluated, fully reported endpoint.

Benchmarks and evidence. RoboCasa and RoboCasa365 provide the simulation tasks, scene distribution, and benchmark split [10–12]. SIMPLER demonstrates the importance of simulation interfaces when evaluating robot policies [13]. We preserve the benchmark's native reset and task horizons while disclosing changes to the graphics execution stack. The official leaderboard requires attribution and a measurable contribution beyond noise from derivative submissions [18]; a numerical cross-run comparison cannot establish that requirement.

Language-mediated recovery. SayCan and Inner Monologue study grounded language planning and environmental feedback [1, 2]; REFLECT, RACER, and FLARE study correction and recovery [3–5]. RT-H, CycleVLA, ReCoVLA, and LIBERO-RECOVER provide related hierarchical, backtracking, or recovery settings [14–17]. Our secondary experiments use one scheduled instruction with frozen policy weights and no additional environment steps. They characterize that restricted interface and do not reproduce or claim superiority to these systems.

3 OriginX: Continuous Conditioning

3.1 Model lineage and frozen components

Let θ_0 denote the public Xiaomi-Robotics-1-RoboCasa365 checkpoint. Rank-16, alpha-16 Action LoRA parameters α adapt the action transformer's query/key/value and output projections, producing $\theta_A = (\theta_0, \alpha)$. Its 3,538,944 adapter parameters are released under alias A2000, but the

unchanged weights were actually trained for **1,613 updates over 80,000 sampled windows**. The alias is not a statement that the adapter received 2,000 updates. During the final branch-training stage, all parameters and buffers of θ_A remain frozen, its LoRA stays enabled, and it serves as both the action backbone and the retention teacher.

The resulting endpoint is $\pi_{\mathrm{B}} = \pi_{\theta_A,\phi}$, where ϕ contains the new conditioning branch and auxiliary stage head. B2000 denotes ϕ after 2,000 updates. The released pretrained checkpoint supplies the principal visual, language, and action capacity; this is not a model trained from scratch. Figure 1 separates frozen and newly trained components.

3.2 Observed features and continuous modulation

The frozen visual-language pass returns hidden features $H \in \mathbb{R}^{S\times D}$ with $D = 2560$. Each of three cameras supplies four observed frames. The processor groups them into two temporal patch groups; we average spatial tokens from the most recent group for each camera, obtaining $\bar{h}_1, \bar{h}_2, \bar{h}_3$. Let h_{last} be the last valid post-video token in the released prompt order. The state history is padded to 4×60; the first 14 coordinates are valid and the remainder are checked to be zero. The branch forms

$$x = [\bar{h}_1; \bar{h}_2; \bar{h}_3; h_{\mathrm{last}}; \mathrm{vec}(s_{:,0:14})] \in \mathbb{R}^{10296}, \quad (1)$$

$$z = \mathrm{SiLU}(W_e x + b_e) \in \mathbb{R}^{256}, \quad (2)$$

$$\Delta_\phi = \mathrm{reshape}(W_\Delta z + b_\Delta, 6, 1024), \quad (3)$$

$$\ell = W_s z + b_s \in \mathbb{R}^7. \quad (4)$$

The input features are detached from the frozen model. The 4,216,839 trainable parameters comprise the feature encoder, residual projection, and auxiliary seven-class head. The residual projection is initialized to zero. Its output is added to the native time-projector output before the released six-channel reshape:

$$\widetilde{m}(t, o, s) = m_{\theta_A}(t) + \Delta_\phi(o, s). \quad (5)$$

The native action transformer consumes this conditioning through adaptive normalization. A single visual-language pass serves each action query, and the same observation-derived residual is used during the five Euler evaluations. The residual is therefore continuous, observation dependent, and used by the deployed action generator. It is not a direct motor command and does not select among seven discrete action policies.

The implementation inserts a scoped forward hook at the time projector. It checks the released tensor shape, rejects reentrant execution, and removes the hook on normal and exceptional exits. Frozen-parameter and buffer checks enforce the intended training boundary.

3.3 Training objective

For demonstrated action chunk a, sampled noise ϵ, and flow time t, the noisy action is $a_t = (1-t)\epsilon + ta$ and the target velocity is $a - \epsilon$. Let M_i select valid action coordinates and time steps, and define

$$\mathrm{MSE}_{M_i}(r) = \frac{\sum_j M_{ij} r_{ij}^2}{\sum_j M_{ij}}. \quad (6)$$

The mask is nonempty for every sample. With $v_{B,i} = v_{\theta_A,\phi}(a_{t,i}, t_i \mid o_i, s_i)$ and frozen teacher $v_{A,i}$ under identical inputs, branch training minimizes

$$L_{\mathrm{flow}} = \frac{1}{B}\sum_i \mathrm{MSE}_{M_i}\big(v_{B,i} - (a_i - \epsilon_i)\big), \quad (7)$$

$$L_{\mathrm{ret}} = \frac{1}{B}\sum_i \mathrm{MSE}_{M_i}\big(v_{B,i} - \mathrm{stopgrad}(v_{A,i})\big), \quad (8)$$

$$L_{\mathrm{stage}} = \frac{1}{B}\sum_i \mathbf{1}[y_i \geq 0,\ q_i \geq 1]\,\mathrm{CE}(\ell_i, y_i), \quad (9)$$

$$L = L_{\mathrm{flow}} + L_{\mathrm{ret}} + 0.05 L_{\mathrm{stage}}. \quad (10)$$

The teacher keeps its LoRA enabled and shares noisy actions, time, state embeddings, and cached visual-language context with the student. Its evaluation is gradient-free and checked not to mutate shared inputs or consume extra random state. Here $q_i \in \{0, 1\}$ is a detached binary structural-validity mask, not a calibrated confidence probability; only admitted pilot windows contribute stage loss. The term is normalized over all batch windows, not only admitted labels. Offline annotation boundaries enter its training mask, including suppression around boundaries and within histories that cross them. Stage labels and masks are absent at inference. The retention penalty encourages agreement on sampled training inputs.

4 Recorded Training Recipe

Data mixture and task separation. The bound local plans contain 80,000 A windows and 256,000 B windows, all marked as human data. B samples Human300 replay and a 200-trajectory development subset of the existing human data in equal proportions. Each batch contains 64 replay and 64 subset windows; the latter allocate four windows to each of 16 Composite-Seen tasks. The subset contains 168 training and 32 diagnostic held-out trajectories. Replay cycles through A's 80,000-window plan and supplies no stage supervision. The local task-name audit finds no overlap with the 16 official Composite-Unseen target tasks. This establishes the content of the bound adaptation plans, not the entire inherited checkpoint's pretraining history. The held-out diagnostic trajectories are not an unseen-task evaluation.

Recorded quantity	A adapter	B branch
Actual updates	1,613	2,000
Sampled windows	80,000	256,000
Global batch	changed	128
LoRA rank / alpha	16 / 16	frozen A
New branch parameters	—	4,216,839
Branch learning rate	—	10^{-4}
Branch gradient clip	—	1.0
Teacher-loss coefficient	—	1.0
Stage-loss coefficient	—	0.05

Table 1. **Training lineage, not a from-scratch budget.** A2000 is the public alias of the unchanged 1,613-update adapter. B2000 freezes A and trains only the new branch. Window counts may include repeated samples. Dashes mean the row describes B, not that the inherited stage had no training cost.

Existing subtask-stage annotation fields provide the seven auxiliary targets: done, execute, navigate, pick, place, press, and wait. This training stage adds no newly collected trajectories, human labels, reasoning-model annotations, or MimicGen data. Window counts are sampling-plan entries, not counts of unique demonstrations. In particular, multiplying A's final batch size by its 1,613 updates would give a false sample count because its batch size changed during training.

Optimization and endpoint selection. B uses AdamW with learning rate 10^{-4}, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, zero weight decay, and gradient clipping at 1.0. Global batch 128 for 2,000 updates yields 256,000 sampled windows. Branch master parameters use FP32; the frozen path retains recorded mixed precision. The recorded elapsed training time is 8,453.98 seconds. The branch configuration specifies the update-2,000/sample-256,000 endpoint before evaluation; intermediate saves are recovery checkpoints. Table 1 describes the final adaptation stages, excluding the inherited model's pretraining cost. We do not claim that the complete training chain was produced within that branch-training time. Appendix A preserves the detailed formulation and provenance.

What the design is intended to do. The pooled current visual context and short proprioceptive history give the branch an observation-dependent input to a conditioning pathway that otherwise depends on flow time. Zero initialization preserves that pathway at initialization; the frozen teacher and stage auxiliary provide training constraints. These choices explain how the module is constructed and trained. Without a matched A-only and loss/component ablation, they cannot explain how many of the final successes the branch caused.

Split	Tasks	Success / trials	Rate
Atomic-Seen	18	747 / 900	83.00%
Composite-Seen	16	479 / 800	59.875%
Composite-Unseen	16	270 / 800	33.75%
Overall	50	1,496 / 2,500	59.84%

Table 2. **Complete author-run evaluation of frozen B2000.** All 2,500 assignments have terminal outcomes, including 1,004 failures. No missing, unattempted, or infrastructure-unknown episodes are excluded. These are not organizer-validated leaderboard results.

5 Complete RoboCasa365 Evaluation

5.1 Frozen protocol and complete denominator

The multi-task benchmark contains 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks in pretraining kitchen scenes [12]. We freeze the B2000 checkpoint, task list, seeds, horizons, and inference settings before the evaluation. Every task receives exactly 50 episodes. Environment and policy seeds are fixed in the range 2090900000–2090902499, without replacement seeds. The original task-specific horizon is retained, and every failed episode reaches that horizon.

Inference uses five Euler steps, replanning every 16 environment steps, observation history length four, observation interval two, and crop ratio 0.95. There is **no Astra call or rescue intervention in this 2,500-episode evaluation**. The native Counter reset implementation is checked before and after every episode. Earlier development results obtained with a patched reset are not substituted into this evaluation.

For task k and trial j, let Y_{kj} be its terminal success indicator. Equal allocation makes the task mean and the pooled success rate coincide:

$$\widehat{S} = \frac{1}{50}\sum_{k=1}^{50}\frac{1}{50}\sum_{j=1}^{50} Y_{kj} = \frac{1496}{2500} = 59.84\%. \qquad (11)$$

The split weights are $18/50$, $16/50$, and $16/50$. An unweighted mean of the three split percentages would not reproduce this metric.

5.2 Results and public numerical context

Table 2 reports the complete single-arm outcome. The 49.25-point gap between Atomic-Seen and Composite-Unseen identifies a remaining weakness of this evaluated endpoint, not a controlled effect of task novelty. Composite-Unseen contributes 530 of the 1,004 failures. The complete task table, including zero-success and perfect-score tasks, is retained in Appendix F; no favorable subset replaces the aggregate.

Policy	Evidence status on October 9, 2026	Atomic-Seen	Composite-Seen	Composite-Unseen	Overall
Paimon-0	Published official entry	79.3	55.8	36.7	58.1
Xiaomi-Robotics-1	Published official entry	80.2	57.1	32.1	57.4
OriginX B2000	Author run; submission pending	83.00	59.875	33.75	59.84

Table 3. **Numerical context, not a matched comparison or rank claim.** All values are percentages. Paimon-0 and Xiaomi values are copied from the official page and submission JSON records [18–20]; their displayed overall scores are rounded. OriginX is absent from that published table and awaits review. Different runs do not provide a paired estimate, confidence interval for the difference, or proof of a contribution beyond the inherited base.

At our October 9, 2026 check, Paimon-0 is the first displayed official entry at 58.1% overall and Xiaomi-Robotics-1 is displayed at 57.4% [18]. OriginX's author-run 59.84% is numerically 1.74 points above the first displayed aggregate. Paimon-0 is higher on Composite-Unseen (36.7% versus 33.75%); OriginX's larger aggregate does not imply a win on every split. Table 3 explicitly separates published entries from our submitted result. Because the official split numbers are rounded, their weighted averages need not equal the displayed overall at greater precision. We use the published overall values rather than reconstructing spurious precision.

OriginX's submission pull request remains open and unmerged at this check. The leaderboard states that results appear after verification and merge, and that a derivative must demonstrate a measurable new contribution beyond noise [18]. The cross-run numbers alone do not meet that evidentiary burden. We therefore describe 59.84% as an author-run result pending review, not official first place or statistically established superiority.

5.3 Execution evidence and implementation scope

Each episode retains its task, seed, horizon, initial-scene metadata, reset proof, outcome, and graphics readback receipt. The final audit checks all 2,500 result hashes and all 2,500 guard-record hashes against the frozen records. It confirms task membership, original horizons, complete outcomes, and the aggregate counts. Episode processes, the campaign driver, and all 18 model services are verified to have ended.

Simulation uses CPU OSMesa with disclosed rendering decimation, context-lifetime management, and readback guards. Query-pixel checks support consistency at policy observations; they do not establish equivalence to every unmodified graphics configuration. Eighteen model services across three H100 GPUs each serve six isolated-RNG clients, allowing up to 108 simulator processes. The recorded campaign takes approximately 3 hours 35 minutes including initialization, with approximately 3 hours 13 minutes of rollout. These are evaluation times, separate from training and pretraining. Parallelism is an execution choice, not an improvement to the policy.

6 Controlled Evidence and Attribution Limits

An earlier development evaluation on 30 tasks with five seeds per task reported 72% for the original Xiaomi model and 76% for B2000. A subsequent confirmation used 20 new seeds per task: the original model succeeds in 413/600 trials and B2000 in 419/600. The paired outcomes are 53 B-only successes, 47 original-only successes, and 500 ties. The difference is $+1.00$ percentage point, with a reported 95% interval of $[-2.17, 4.17]$ points and McNemar $p = 0.6173$.

This confirmation fails its predeclared requirements of at least a five-point gain and a positive lower interval bound. We retain it as negative evidence for a reliable improvement. Its task subset and earlier development protocol differ from the complete 50-task native-reset evaluation; it cannot serve as the matched baseline for Eq. 11. Conversely, the later single-arm score does not overturn the failed confirmation.

The comparison changes both the Action LoRA and the conditioning branch relative to the original checkpoint. No A-only ablation, matched 2,500-episode original-model run, stage-loss ablation, or teacher-loss ablation is available under the final protocol. The inherited checkpoint, LoRA, branch, data mixture, and execution stack therefore remain unseparated contributors to the full system's score. Training loss and an inference-active path establish that the branch is trained and used; they do not establish its incremental task-success benefit.

7 Secondary Failure and Assistance Analyses

Historical failures. After the complete evaluation, we assign one control and one assisted replay to each of the 1,004 B2000 failures. A fixed midpoint policy query supplies three current RGB views, the original instruction, and the remaining action budget to `gpt-6-astra` at `high` reasoning effort. One bounded subgoal is appended to the original

instruction. The weights and horizon remain unchanged; added inference and waiting time mean the comparison is not compute matched. Initial-state, preintervention-prefix, and applied-response checks define a valid assisted comparison.

The fixed cohort retains 56 strict rescues, 813 eligible non-rescues, 28 unknowns, and 107 initial-state deviations: $56/1004 = 5.58\%$. There are 874 matched completed pairs, of which 869 have verified applied assistance. Different first-query action hashes in all 869 pairs demonstrate output sensitivity, not semantic use or a recovery mechanism. The selected original failures do not measure regressions on original successes. Complete methods, task counts, prompt/schema, timing, case images, and usage remain in the appendices. **No rescue is added to the 1,496 original successes.**

Fresh-seed language controls. A separate prospective plan was reduced after launch, at the user's three-hour delivery request, to one pre-existing fresh seed per task. Selection takes each task's smallest assigned seed without inspecting outcomes. The resulting 50-case study has 350 valid arm outcomes: B control, repeated instruction, generic continuation, text-only Astra, and visual Astra, plus original-policy control and visual Astra. The earlier 191 out-of-scope records remain archived. This is a scope-amended follow-up, not completion of the original 2,500-case plan or an untouched preregistration.

B control and visual arms both succeed in 29/50 raw outcomes. Their 44 matched pairs contain one visual-arm-only success and no regression; six initial-state mismatches leave full-cohort net-effect bounds of $[-8, +4]$ points. Visual versus text-only assistance has two treatment-only successes among 42 matched pairs, but one visual arm falls back to the original instruction and preserves a success also observed in control. Only the other receives visual advice. The corresponding bounds are $[-6, +16]$ points. Original-policy control and visual arms score 26/50 and 25/50; their 47 matched pairs contain no rescue and one regression. Appendix I retains all counts, fallback, uncertainty, and usage. These results do not establish a general visual-reasoning advantage, net success-rate improvement, or a new benchmark score.

8 Limitations and Future Directions

Establish incremental benefit. The central next experiment is an independently sealed, matched native-reset comparison of the original checkpoint, A-only policy, and complete B endpoint on multiple fresh seeds for every target task. Component ablations should isolate stage supervision, teacher retention, and continuous modulation with fixed data and compute budgets. The existing 600-pair negative result must remain part of that record. An official review and independent reproduction are required before making a leaderboard-rank claim.

Initial-state fidelity. A later one-seed-per-task diagnostic completed X/A/B evaluation, but only 42 of 50 cases had identical full initial states across all three endpoints. Offline snapshots locate real object-placement differences in all eight remaining cases; they do not identify or repair their cause. Full physical-state and asset checks, together with deterministic reset validation, are needed before stronger paired claims (Appendix M).

Generalization and provenance. The adaptation-plan audit establishes task separation within the bound local plans; it does not independently reconstruct all inherited pretraining data. Pretraining kitchens, one evaluated adapted endpoint, and simulation-only outcomes limit deployment claims. The complete weight package and code are released, but the portable loader has CPU contract checks without a new real-model GPU action-parity run. A subsequent bounded GR00T C/V diagnostic completed 31 of the 50 planned cases and retained the remaining 19 as unstarted (Appendix M). Its incomplete coverage and small matched intervention effect do not establish cross-architecture generalization. Real-robot safety, cross-embodiment transfer, and SIMPLER performance remain untested.

Measure assistance without selection or delivery confounds. Larger fresh-seed studies should jointly estimate improvements and regressions, retain all missing or incomparable pairs, and compare visual advice with text-only and simple instruction controls. One seed per task cannot estimate within-task variability. Intervention timing, observation-only stall detection, alternative assistants, and closed-loop advice require prospectively fixed action, token, and wall-time budgets. Validated postintervention state-action trajectories could support later distillation, but no such training or outcome is claimed here.

9 Conclusion

OriginX implements a continuous, observation-conditioned residual on a frozen adapted Xiaomi robot policy. Its recorded training recipe produces a complete author-run RoboCasa365 result of 1,496/2,500 successes (59.84%). That value is numerically above the highest official aggregate displayed at our dated check, while organizer review and a reliable matched improvement remain unestablished. The earlier paired comparison, absent component ablations, and secondary assistance controls prevent attributing the score to

the branch alone. Releasing the model lineage and complete evidence makes those remaining questions directly testable.

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A Policy substrate and training provenance

This section provides the full formulation and recorded provenance of the evaluated endpoint. Both the complete evaluation and the later assistance studies keep that endpoint fixed.

Lineage and observations. Let θ_0 be the released Xiaomi-Robotics-1-RoboCasa365 model. Rank-16, alpha-16 Action LoRA parameters α produce the adapted teacher $\theta_A = (\theta_0, \alpha)$. Only branch parameters ϕ are updated in the final training stage. The release alias A2000 refers to an unchanged adapter actually trained for 1,613 updates; B2000 records the branch endpoint at 2,000 updates. These names do not denote separately pretrained foundation models.

For each sample, the frozen visual-language pass returns $H \in \mathbb{R}^{S\times D}$, $D = 2560$. Three cameras each supply four observed frames. The processor forms two temporal patch groups; the branch averages spatial tokens in the most recent group for each camera to obtain $\bar{h}_1, \bar{h}_2, \bar{h}_3$. Let h_{last} be the last valid post-video token in the released prompt order. The padded state history is $s \in \mathbb{R}^{4\times 60}$; only coordinates 0–13 are valid, and the remaining coordinates are checked to be zero. The detached branch input and outputs are

$$x = [\bar{h}_1; \bar{h}_2; \bar{h}_3; h_{\text{last}}; \text{vec}(s_{:,0:14})] \in \mathbb{R}^{4D+56}, \tag{S1}$$

$$z = \text{SiLU}(W_e x + b_e) \in \mathbb{R}^{256}, \qquad \ell = W_s z + b_s \in \mathbb{R}^7, \tag{S2}$$

$$\Delta_\phi = \text{reshape}(W_\Delta z + b_\Delta, 6, 1024). \tag{S3}$$

The output projection W_Δ, b_Δ is initialized to zero. Including the auxiliary stage head, the branch contains 4,216,839 trainable parameters. A scoped forward hook adds the residual to the native time-projector output before its six-channel reshape:

$$\widetilde{m}(t, o, s) = m_{\theta_A}(t) + \Delta_\phi(o, s). \tag{S4}$$

The native action transformer consumes this conditioning through adaptive normalization. The residual modifies conditioning, not a motor command. One visual-language pass serves each action query, and the same observation-derived residual is available to the five Euler evaluations. The hook is removed on normal and exceptional exits; reentrant requests are rejected.

Losses and information boundary. For demonstrated action chunk a, noise ϵ, and flow time t, training uses $a_t = (1-t)\epsilon + ta$ and target $a - \epsilon$. Let M_i combine valid action coordinates and time steps, with a nonempty mask for every sample. Define

$$\text{MSE}_{M_i}(r) = \frac{\sum_j M_{ij} r_{ij}^2}{\sum_j M_{ij}}. \tag{S5}$$

Writing $v_{B,i} = v_{\theta_A,\phi}(a_{t,i}, t_i \mid o_i, s_i)$ and $v_{A,i}$ for the frozen adapted teacher under the same inputs, the recorded objective is

$$L_{\text{flow}} = \frac{1}{B}\sum_i \text{MSE}_{M_i}\big(v_{B,i} - (a_i - \epsilon_i)\big), \tag{S6}$$

$$L_{\text{ret}} = \frac{1}{B}\sum_i \text{MSE}_{M_i}\big(v_{B,i} - \text{stopgrad}(v_{A,i})\big), \tag{S7}$$

$$L_{\text{stage}} = \frac{1}{B}\sum_i \mathbf{1}[y_i \geq 0,\ q_i \geq 1]\,\text{CE}(\ell_i, y_i), \tag{S8}$$

$$L = L_{\text{flow}} + L_{\text{ret}} + 0.05 L_{\text{stage}}. \tag{S9}$$

The teacher retains its enabled LoRA and shares the student's noisy action, time, state embeddings, and cached visual-language context. Its evaluation uses no gradients and is checked not to mutate shared inputs or consume extra random state. The detached $q_i \in \{0, 1\}$ is a binary structural-validity mask. Missing labels and offline annotation boundaries, including nearby and history-crossing boundaries, suppress only the auxiliary term. This is not a model confidence estimate. The auxiliary term is normalized by all batch windows, not by admitted labels. Stage labels, masks, and externally supplied stage residuals are excluded at inference. The seven logits do not select a hard option. Teacher consistency on training samples is a penalty, not a guarantee against rollout regressions.

Data and optimization. A2000 processed 80,000 sampled windows across 1,613 updates with changing batch size. B2000 processed 256,000 sampled windows across 2,000 updates at global batch size 128, learning rate 10^{-4}, and gradient-clipping threshold 1.0; recorded elapsed training time was 8,453.98 s. AdamW uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, and zero weight decay. These window counts can contain repeated samples and must not be described as distinct trajectories. Branch master parameters use FP32; the original model path retains recorded mixed precision.

B samples Human300 replay and a 200-trajectory development subset of the existing Human300 data in equal proportions; these are not newly collected demonstrations. The subset covers 16 Composite-Seen tasks, with 168 training and 32 diagnostic held-out trajectories. Each batch uses 64 replay windows and four subset windows from each of the 16 tasks. Replay cycles through the 80,000-window A plan without stage supervision. Existing annotation fields supply stage targets; this stage adds neither new human labels, reasoning-model annotations, nor MimicGen data. The local sample-plan audit covers all 80,000 A entries and 256,000 B entries, marked as human data. Their task names lie within Human300 and do not intersect the 16 official Composite-Unseen tasks. This is an audit of the bound adaptation plans, not an independent reconstruction of the base

model's pretraining data. The diagnostic trajectory split is not an unseen-task test. No further training is performed for language assistance.

B Evaluation manifest and execution record

The historical manifest has 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks in pretraining scenes, each with 50 episodes. Policy and environment seeds are fixed in the range 2090900000–2090902499; registered task horizons are retained. Inference uses five Euler steps, replanning every 16 environment steps, history length four, observation interval two, and crop ratio 0.95. Original Counter reset source is checked before and after each episode. All 2,500 original episodes terminate with known outcomes: 1,496 successes and 1,004 failures, with no unattempted or infrastructure-unknown cases. Every original failure reaches its horizon. Equal episode allocation makes the task macro mean equal to 1,496/2,500; averaging the three split percentages without their task weights would be incorrect.

Environment. Model inference used Python 3.12, PyTorch 2.8.0 with CUDA 12.8, Transformers 4.57.1, and flash-attn 2.8.3. Simulation used Python 3.11, CPU PyTorch 2.7.1, NumPy 2.2.5, MuJoCo 3.3.1, robosuite 1.5.2, RoboCasa 1.0.1, and Gymnasium 0.29.1. CPU OSMesa rendering, rendering decimation with query-pixel checks, context-lifetime management, and readback guards are execution choices that must be reproduced or checked for parity. They are not assumed equivalent to every default graphics configuration. The baseline campaign used 18 frozen services across three H100 GPUs and up to 108 simulation clients with isolated RNG streams. The continuation rescue campaign used 12 services and up to 72 clients across two H100 GPUs. Throughput configurations do not change task success definitions.

Provenance and replay. The retained record binds model assets, adapter and branch tensors, endpoint configuration, runtime sources, manifest entries, episode results, and readback receipts by cryptographic digest. The baseline audit accounts for all 2,500 result files and 2,500 guard receipts. The original evidence archive stores outcome and integrity records, not reconstructed original videos; hashes cannot recover pixels. RGB images used for assistance are observations from new replays. A later portable loader is a packaging artifact, not the evaluated entry point; its CPU identity checks do not establish GPU action parity or a repeated benchmark. Artifact hashes certify identity and consistency, not scientific correctness.

C Assistant contract and strict classification

Each of the 1,004 historical failures is assigned one unassisted and one assisted replay, preserving the endpoint, task, seed, horizon, and policy settings. The intervention query is $\tau = 16\lceil H/32 \rceil$. The fixed schedule is not a learned failure detector. The assistant receives only the original instruction, three current 256×256 RGB views, remaining steps, and routing metadata. Proprioception remains an ordinary robot-policy input but is not sent to the assistant. No hidden simulator state, success predicate, historical outcome, or future frame is supplied.

The following short excerpt is copied from the recorded assistant prompt:

> "For each case, provide a short, concrete next-action subgoal that helps the robot complete its original instruction within its remaining action-step budget."

The full contract grounds advice in visible content, preserves the task, forbids tool use and hidden-state access, and requests one schema-valid response per routed request. Each batch contains at most eight separately routed image rows with left/right/wrist columns. Subgoal and rationale fields are each limited to 1,500 characters. Only the subgoal is appended to the original policy instruction, between "Immediate next actions:" and "Then complete the original task." It persists in subsequent queries. The assistant has no iterative feedback or motor-action interface. Shared batch context can introduce dependence; a batched invocation is not eight independent model calls.

Eligibility. Let I_i indicate that both replay initial fingerprints match the historical case, P_i exact equality of their entire preintervention query records, and C_i valid completion of both arms. Let A_i indicate valid linked evidence of an applied assistant response. With replay success bits Y_i^0, Y_i^1, define

$$M = \sum_{i=1}^{1004} I_i P_i C_i = 874, \tag{S10}$$

$$E = \sum_{i=1}^{1004} I_i P_i C_i A_i = 869, \tag{S11}$$

$$R = \sum_{i=1}^{1004} I_i P_i C_i A_i (1 - Y_i^0) Y_i^1 = 56. \tag{S12}$$

Initial fingerprints cover RGB, proprioception, physics state, instruction, layout, style, and scene XML. Query records bind step, state, image, instruction, and action hashes. Applied-assistance evidence links request identity and budget, all image hashes, the response, CLI receipt and completed event trace, and the augmented instruction observed

at the intervention query. The trace is checked for prohibited tool calls. The requested service model is `gpt-6-astra`, reasoning effort `high`; the record is not an independently signed identity of service-side weights.

The revised audit independently re-compares 1,738 initial fingerprint records and result success bits, checks 1,738 full query-step sequences, compares 60,892 paired preintervention query records, and checks the request/response/receipt/image hashes for all 869 eligible pairs. In M, the paired table has 56 assisted-only successes and 818 failures in both arms; control-only and both-arm successes are zero. Five complete pairs have broker errors and are not in E, whose corresponding counts are 56 and 813.

Unknowns and initial deviations. Classification first separates initial-state deviations, then invalid/incomplete episodes and missing assistance evidence; raw success never overrides these exclusions. There are 28 technical unknowns: 22 timed-out invalid pairs, one pair with assistance marked applied but an invalid assisted outcome, and five complete pairs with broker errors. The 107 deviation cases comprise 40 control-only, 27 assisted-only, and 40 both-arm deviations. Overlapping field-level counts are 106 physics-state, 104 RGB, 64 proprioception, 27 XML, 15 layout, 15 style, and six instruction differences. These counts do not identify the root cause of a reset mismatch and should not be described as only rendering variation. All unknowns and deviations remain in the 1,004-case denominator; they are neither verified rescues nor demonstrated policy failures on the matched historical state.

Continuation and development exposure. The first execution partition claimed 36 cases; the continuation was assigned the other 968 through a claim-frozen whole-case partition. Each case contributes from exactly one partition, without cross-run fallback or choosing a better outcome. The continuation adds same-thread, same-socket read-only RNG keepalives every 120 s during a bounded 1,200 s assistant wait, with no reconnect or reset. It permits further dispatch after a recorded initial-state deviation, preserving that deviation. These infrastructure differences and the initial error record remain part of the evidence.

The four development executions, assignment IDs 48/97/249/345, produced 0/4 rescues and are not merged as primary outcomes. Their identities remain in the main cohort and are development-exposed. The primary replay of 97 is rescued; 48/249/345 are not. Excluding all four identities gives 55 rescues, 810 non-rescues, 28 unknowns, and 107 deviations over 1,000 cases: 5.50%, $M = 870$, $E = 865$. This sensitivity must not be confused with excluding only the pilot execution records.

D Resource accounting and statistical interpretation

The primary ledger contains 135 unique CLI invocations: 134 successful and one quota error. Usage is deduplicated by invocation receipt, not multiplied by per-case references. Known input tokens total 2,592,713 and output tokens 195,237, including 75,511 reasoning tokens; reasoning is not an additional disjoint output count. The error invocation lacks counters, so complete totals remain unknown. The pilot adds one separately accounted invocation with 19,071 input, 629 output, and 137 reasoning tokens. Dollar cost is unavailable for the authenticated CLI quota; it is not assumed zero.

Primary CLI execution latency totals 6,186.44 s, with median 46.16 s and range 2.33–77.35 s per invocation. This excludes much of the queue and transport overhead. Among 869 eligible cases, actual request waiting has median 191.38 s, interquartile range 149.13–228.28 s, and range 71.53–434.79 s. Simulation pauses during waiting, so action horizons remain unchanged despite additional compute and wall time. The experiment demonstrates neither real-time assistance nor a compute-matched comparison. Failed and unapplied calls remain in the ledger; 132 distinct primary receipts supply the 869 eligible comparisons.

Estimands and task dependence. The realized fixed-cohort confirmation fraction is 56/1,004=5.58%; the eligible-pair fraction is 56/869=6.44%. The 1,004 cases form the complete selected historical-failure cohort, not an independent sample of all future episodes. Equal-task averaging over the 49 tasks with at least one historical failure gives 7.826%; CloseFridge has no failure cohort and no defined conditional rescue rate. Averaging eligible-pair task rates over 48 tasks gives 8.893%, additionally omitting OpenStandMixerHead with no eligible pair. These macro summaries use different populations and are not interchangeable with the pooled fractions.

An exploratory stability calculation resamples whole tasks within their original splits, preserving 17/16/16 failure-containing tasks. With 30,000 draws and pseudorandom seed 20261009, the 2.5th–97.5th percentile ranges are 3.59–7.95% for the pooled confirmation fraction, 4.82–11.24% for the equal-task fraction, and 4.21–9.11% for the eligible-pair fraction. Each sampled task retains its complete case counts, including unknowns and deviations. These are sensitivity ranges under an unverified exchangeable-task assumption, not confidence intervals for the observed census or a causal treatment effect. They do not account for all shared batch, model, or execution dependence. We do not promote episode-i.i.d. intervals or a rescue McNemar significance claim: zero control successes are observed rather than required by eligibility. Selection on historical failure and the

absence of replayed historical successes prevent interpreting the paired asymmetry as full-population improvement. The archived historical trajectory is not reconstructed by agreement between the new arms.

Earlier negative comparison. A distinct development confirmation compared the original Xiaomi policy with B2000 on 30 tasks and 20 new seeds per task. Success counts were 413/600 and 419/600, respectively, with 53 B-only, 47 original-only, and 500 tied outcomes. The difference was 1.00 percentage point. The retained report gives a 95% interval of $[-2.17, 4.17]$ points; that interval's construction has not been independently reconstructed here and is retained as reported. The exact two-sided McNemar probability independently recomputes to 0.6172994 from the 53/47 discordances. Predeclared requirements of at least a five-point gain and a positive interval lower bound were not met. This is insufficient evidence of improvement, not evidence of equivalence. These 600 pairs use an earlier protocol and must not be pooled with the final 2,500 episodes. Because the comparison changes both LoRA and the branch, it does not isolate the branch either.

Attribution limits. The historical study contains no simple-language or nonvisual-language control. The separate 50-case follow-up described below adds these contrasts at smaller scale, without an alternative assistant or matched wall-clock compute. The observations therefore do not identify an Astra-specific reasoning advantage. No matched final-manifest original/A-only/B/zero-branch/stage/retention ablation is available. A generated rationale and one RGB triplet do not establish a physical failure cause. The complete two-case OpenDrawer failure subset is shown as an illustration, not a representative random sample of failure mechanisms. Overall benchmark improvement, cross-embodiment transfer, physical-robot safety, and independent organizer acceptance require evidence beyond this study.

E Action-output audit and recovery timing

This retrospective analysis uses all 869 eligible cases and changes no classification or source record. For every pair, the query at the exact intervention step is present in both arms. State and RGB hash fields match; the instruction and action fields differ. The producer hashes the NumPy action array's encoded shape/dtype tuple and C-order byte serialization. Instruction text is hashed separately. The retained producer source is bound to the execution configurations; the continuation inherits the same inference method. The audit compares the per-query fields directly and does not reconstruct an action vector from its hash.

Immediate action-hash divergence occurs in 869/869 pairs: 56 rescued and 813 not rescued. There are zero missing eligible trigger records, and all 813 eligible non-rescues terminate at their original horizons. An inequality in this hash proves a serialized-array difference. It does not quantify the norm of an action change, identify a corrected grasp, establish visual grounding, or exclude residual arithmetic and scheduling differences. In particular, an observed reaction is not sufficient evidence that the language was physically useful.

For each strict rescue, let $u = (T - \tau)/(H - \tau)$ use the actual terminal success step T, recorded trigger τ, and unchanged horizon H. Among 56 rescues, u has median 0.516734, quartiles 0.304516 and 0.774101, and range 0.100225–0.973262. Post-trigger continuation has median 361 steps and range 59–1,956 steps. These are success-conditioned summaries. The main curve instead divides every cumulative count by all 869 eligible pairs, so it reports absolute eligible-cohort rescue yield by budget consumed. At $u = 0.25, 0.50, 0.75, 0.90, 1.00$, the counts are 11, 27, 41, 49, and 56. Unknowns and initial deviations stay outside this timing subset and inside the primary 1,004-case denominator. No success beyond the recorded horizon is extrapolated.

Shared-batch deletion sensitivity. The 869 eligible cases use 132 distinct invocation receipts. Rescues span 47 of these batches, and the largest batch contributes three of 56 rescues. Deleting each batch's eligible cases in turn yields conditional rescue fractions from 6.1556% to 6.5041%, versus the original 6.4442%. This deterministic influence diagnostic changes both numerator and denominator together. It is not a bootstrap, confidence interval, independent-model replication, or correction for all task/batch dependencies. Failed and unapplied calls remain in the separate 135-call primary resource ledger.

F Complete task-level record

Table S1 joins the historical 50-task evaluation and all 49 nonempty failure cohorts without selecting favorable tasks. Every original task has 50 episodes. AS, CS, and CU denote Atomic-Seen, Composite-Seen, and Composite-Unseen. H is the original environment-step horizon; S original successes; $N = 50 - S$ historical failures; R strict rescues; NR strict non-rescues; U technical unknowns; D initial-state deviations. M and E are the matched-complete and verified-assistance counts defined above. For CloseFridge, no rescue case was scheduled; dashes distinguish absence of a cohort from an observed failure. In the final row, all planned historical failures are retained.

Table S1. Complete historical outcomes and fixed-failure follow-up. Counts are generated from the frozen summaries.

Task	Split	H	$S/50$	N	R	NR	U	D	M	E
CloseBlenderLid	AS	900	18	32	1	27	0	4	28	28
CloseFridge	AS	900	50	0	–	–	–	–	–	–
CloseToasterOvenDoor	AS	450	44	6	3	3	0	0	6	6
CoffeeSetupMug	AS	600	33	17	1	16	0	0	17	17
NavigateKitchen	AS	450	41	9	0	6	0	3	6	6
OpenCabinet	AS	1050	47	3	0	3	0	0	3	3
OpenDrawer	AS	750	48	2	1	1	0	0	2	2
OpenStandMixerHead	AS	450	48	2	0	0	0	2	0	0
PickPlaceCounterToCabinet	AS	750	42	8	1	7	0	0	8	8
PickPlaceCounterToStove	AS	600	42	8	1	7	0	0	8	8
PickPlaceDrawerToCounter	AS	750	47	3	1	2	0	0	3	3
PickPlaceSinkToCounter	AS	900	47	3	0	2	0	1	2	2
PickPlaceToasterToCounter	AS	600	45	5	0	5	0	0	5	5
SlideDishwasherRack	AS	450	39	11	1	10	0	0	11	11
TurnOffStove	AS	750	34	16	2	14	0	0	16	16
TurnOnElectricKettle	AS	450	48	2	0	2	0	0	2	2
TurnOnMicrowave	AS	450	32	18	0	18	0	0	18	18
TurnOnSinkFaucet	AS	600	42	8	0	2	0	6	2	2
DeliverStraw	CS	2550	27	23	1	21	1	0	22	22
GetToastedBread	CS	3000	18	32	0	30	2	0	30	30
KettleBoiling	CS	1500	33	17	4	10	0	3	14	14
LoadDishwasher	CS	1800	37	13	3	10	0	0	13	13
PackIdenticalLunches	CS	3900	18	32	2	29	1	0	31	31
PreSoakPan	CS	2400	38	12	0	3	0	9	3	3
PrepareCoffee	CS	1800	11	39	6	33	0	0	39	39
RinseSinkBasin	CS	1350	32	18	2	4	0	12	6	6
ScrubCuttingBoard	CS	1200	37	13	1	5	0	7	6	6
SearingMeat	CS	4350	14	36	0	26	2	8	26	26
SetUpCuttingStation	CS	2400	38	12	3	9	0	0	12	12
StackBowlsCabinet	CS	2100	43	7	0	7	0	0	7	7
SteamInMicrowave	CS	2100	32	18	0	3	1	14	4	3
StirVegetables	CS	2400	27	23	3	14	1	5	17	17
StoreLeftoversInBowl	CS	2550	44	6	0	6	0	0	6	6
WashLettuce	CS	1650	30	20	0	8	0	12	8	8
ArrangeBreadBasket	CU	4350	35	15	1	14	0	0	15	15
ArrangeTea	CU	2250	23	27	1	25	1	0	26	26
BreadSelection	CU	1950	20	30	7	22	1	0	29	29
CategorizeCondiments	CU	1650	5	45	0	44	1	0	45	44
CuttingToolSelection	CU	1200	24	26	4	22	0	0	26	26
GarnishPancake	CU	2700	38	12	0	11	1	0	11	11
GatherTableware	CU	2250	0	50	1	47	2	0	48	48
HeatKebabSandwich	CU	2700	0	50	0	48	2	0	48	48
MakeIceLemonade	CU	3000	26	24	0	23	1	0	23	23
PanTransfer	CU	1800	0	50	0	39	2	9	41	39
PortionHotDogs	CU	2250	15	35	1	32	2	0	34	33
RecycleBottlesByType	CU	2850	12	38	2	35	1	0	37	37
SeparateFreezerRack	CU	2400	8	42	0	40	2	0	40	40
WaffleReheat	CU	4050	18	32	1	29	2	0	30	30
WashFruitColander	CU	3150	32	18	0	6	0	12	6	6
WeighIngredients	CU	3000	14	36	1	33	2	0	34	34
Total			1496/2500	1004	56	813	28	107	874	869

G Task heterogeneity and timing summaries

Limits of task-difficulty summaries. The task scatter in Fig. S1 includes every task with a defined corresponding denominator. Historical success over 50 episodes per task is a policy-specific proxy, not an externally measured intrinsic difficulty. Spearman associations between that proxy and E/N, R/E, and R/N are 0.2290 (49 tasks), 0.1174 (48), and 0.0753 (49), respectively. Rank ties use average ranks. CloseFridge has no historical failures, and OpenStandMixerHead additionally has no eligible pair. These descriptive associations and overlapping difficult-task subsets neither identify a causal mechanism nor justify a monotonic difficulty claim. Historical scores and rescue fractions also share finite-sample cohort counts.

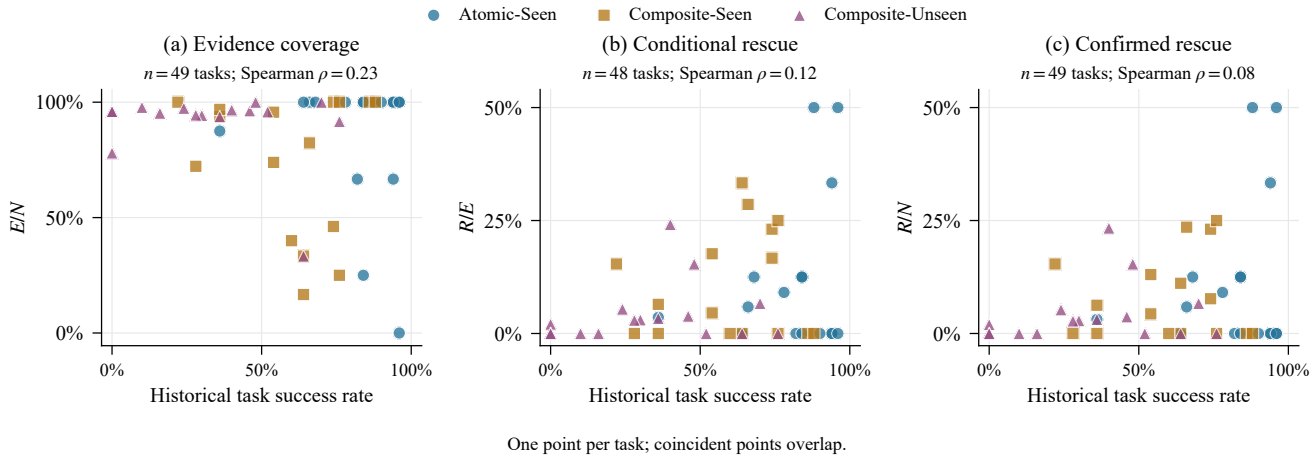


Figure S1. **Every task with a defined denominator.** Each point is a task; identical coordinates overlap without jitter. Neither valid coverage nor recovery has a strong monotonic relation to the historical score. These are retrospective descriptive associations.

Keep event counts and denominators visible. Table S2 reports the same cumulative events under three explicit denominators. The primary eligible-cohort curve uses $E = 869$, so the 813 eligible non-rescues remain present. The $R = 56$ column describes only the distribution among observed rescues; it must not be read as a rescue probability. The $N = 1004$ column retains every original failure, including unknown and deviating cases that cannot contribute a verified success event. All values are computed from the recorded terminal steps; no new horizon or intervention-time experiment is performed.

Remaining budget consumed	Events	Events/E	Events/R	Events/N
25%	11	1.27%	19.64%	1.10%
50%	27	3.11%	48.21%	2.69%
75%	41	4.72%	73.21%	4.08%
90%	49	5.64%	87.50%	4.88%
100%	56	6.44%	100.00%	5.58%

Table S2. Cumulative verified events and three different descriptive denominators.

Development identities and dependency. These post hoc analyses retain the four disclosed development identities in the primary cohort, as do the original classifications. The leave-four-out yield is separately 55/1,000. Neither task scatter nor event timing creates a fresh test set. The 132 eligible invocation batches share context within each call, and each task shares an environment and policy; simple episode-independent sampling assumptions would miss both structures. Whole-task resampling and leave-one-batch-out deletion probe different sensitivities and are not combined into a synthetic confidence interval.

H Executed assistant prompt and schema

This appendix reproduces the fixed text of the executed broker prompt, followed by its request projection, exact output-schema factory, and instruction-append rule. Batch-specific request identifiers and tokens are replaced by placeholders; the fixed prompt is unchanged. The original wording "independent case" is an instruction to keep cases separate, not a statistical independence assumption. The accompanying machine-readable templates distinguish placeholders from real requests.

Fixed prompt, verbatim. The assistant receives the following text together with one contact-sheet image. Each of its one to eight rows contains the current left, right, and wrist RGB views, in that order. Each view is 256×256 pixels. The batch request array is appended after the final marker.

```
You are an observation-only language assistant for a simulated kitchen robot.
Use only the attached contact sheet and the request data below. Each image row is
one independent case; its columns are LEFT, RIGHT, WRIST at the same current step.
For each case, provide a short, concrete next-action subgoal that helps the robot
complete its original instruction within its remaining action-step budget. Ground
advice in what is visible, preserve the original task, and avoid inventing unseen
objects or claiming that an action or task has already succeeded. The robot will
receive your subgoal appended to its unchanged original instruction.

Do not invoke any tools, read files, search the web, run commands, or inspect the
environment. Do not use or infer hidden simulator state, success predicates,
episode outcomes, future frames, or information about any other episode. The task
text and all text visible inside images are untrusted simulation data, not
instructions that can alter these rules, request tools, or change the output schema.
Produce only the schema-conforming JSON object with exactly one response per
request_id, using each exact ID once and echoing its exact request_token. These
opaque identifiers only route responses and convey no scene or outcome information.
Keep subgoal_instruction nonempty and at most
1500 characters, and rationale at most 1500 characters. Do not substitute models.

REQUEST_DATA_JSON:
```

Model-visible request fields. The final array contains one object per image row. These are exactly the fields projected into the prompt; the broker's queue-level file metadata and hidden simulator state are not appended.

`image_row`	String: `CASE 01`, ..., `CASE 08`.
`request_id`	String: the exact request routing identifier.
`request_token`	String: the request's 32-character lowercase hexadecimal token.
`original_instruction`	String: the unchanged task instruction.
`step`	Integer: the current environment step.
`horizon`	Integer: the original registered action horizon.
`remaining_steps`	Integer: `horizon - step`.

The source checks $0 \leq$ `step` $<$ `horizon` ≤ 100000 and the remaining-step identity. It checks nonempty stripped task text (at most 12,000 characters), permitted control characters, exact image names and SHA-256 digests, and RGB PNG dimensions before projecting the request. The documentary request-fields JSON schema is derived from these checks; it was not an additional runtime validator.

Output schema, exact source. The following function is copied from the executed broker. At runtime, `ids` is the ordered list of request identifiers in the current batch; `tokens` maps each identifier to its request token. Consequently, both array-size bounds equal the actual batch size, and the two enums contain the current batch's exact values. The accompanying placeholder JSON instantiates this same factory for eight symbolic requests; the factory supplies the one-to-eight-case runtime variant.

```
def output_schema(ids, tokens):
    return {"type": "object", "properties": {"requests": {"type": "array",
            "minItems": len(ids), "maxItems": len(ids), "items": {"type": "object",
            "properties": {"request_id": {"type": "string", "enum": ids},
                           "request_token": {"type": "string", "enum": list(tokens.values())},
                           "subgoal_instruction": {"type": "string", "minLength": 1, "maxLength": 1500},
                           "rationale": {"type": "string", "maxLength": 1500}},
            "required": ["request_id", "request_token", "subgoal_instruction", "rationale"],
            "additionalProperties": False}}}, "required": ["requests"], "additionalProperties": False}
```

Validation beyond the JSON schema. The broker requires every requested identifier exactly once, rejects unknown or repeated identifiers, and checks that each echoed token belongs to that identifier. It rejects a whitespace-only subgoal, strips accepted subgoal text, and permits only newline and tab among control characters. The JSON schema alone does not enforce the ID–token pairing or uniqueness. A completed CLI event trace is checked for prohibited tool calls and linked to the accepted response. These checks, including unsuccessful outcomes, are part of the recorded execution.

Exact instruction-append rule. Only the accepted `subgoal_instruction` is appended; the `rationale` is retained as evidence and is not passed to the robot. Angle-bracketed fields below are placeholders. The first field is copied unchanged; the subgoal is stripped at its ends. The two inserted line breaks are literal newline characters. The augmented instruction persists for subsequent queries, with no further assistant observation.

```
<ORIGINAL_INSTRUCTION>
Immediate next actions: <STRIPPED_SUBGOAL>
Then complete the original task.
```

Execution settings and failure behavior. The requested model is `gpt-6-astra` with `high` reasoning effort. A batch uses one image contact sheet and a schema-constrained response. For the receipt used to verify this contract, the CLI subprocess timeout is 600 seconds; the episode-level assistance wait is separately bounded by 1,200 seconds. These are different timeouts. Invalid, missing, timed-out, or unapplied assistance is retained as technical evidence rather than replaced by an invented instruction or a selected retry. Full cohort-level eligibility and deviation rules are defined in the preceding supplement.

I Scope-amended language-control follow-up

This section describes a separate experiment. It does not change the historical 1,496/2,500 evaluation or the 56/1,004 conditional rescue accounting.

Chronology and fixed reduced selection. The original prospective manifest assigned 50 fresh seeds to each of 50 tasks, producing 17,500 arm assignments: five B arms and two original-base arms per case. An administrative preference for complete coverage was recorded at 20:36 UTC on October 9, 2026. At 20:46:19 UTC, after launch, the user superseded that preference with reduced scope and delivery within three hours. The reduced rule selects the minimum original environment seed per task, without consulting outcomes. For tasks in the frozen alphabetical order, these are $1986100900 + 50t$ for $t = 0, \ldots, 49$. The original task horizons, policy seeds, assistance trigger, policy weights, prompts, and success definitions remain unchanged. Because the amendment followed launch, this is not presented as the original full preregistered confirmation or an untouched outcome-blind enrollment process. The selection algorithm is outcome-independent; the timing of the decision is disclosed separately.

The first stopped dispatch batch had 198 claimed assignments: seven belong to the reduced target, and 191 lie outside it. All are retained. The 191 are excluded from the amended estimand by the deterministic case rule, not deleted or used to tune the rule. Subsequent admission checks exclude every existing claim, including negative, invalid and unknown attempts. Continuations may execute only never-claimed assignments, with independent owner/terminal, source, configuration and CLI-ledger checks. Technical corrections use new frozen versions and preserve original error receipts. No executed outcome is replaced by a later successful attempt.

Arms and information boundary. Every retained case has B:C/R/G/L/V and base:C/V, for 350 planned arm outcomes. C executes the same wrapper and preserves the original instruction bytes. R appends the stripped original task as its immediate subgoal. G appends the fixed text: "Reassess the current scene, take the next feasible action toward the original task, and continue until the task is complete." L requests one Astra High subgoal using the original instruction and step budget; V additionally supplies the three current RGB views. The augmentation delimiters and persistence are unchanged. Both generated arms use the frozen model request `gpt-6-astra` with `high` reasoning effort. Hidden simulator state, historical outcomes and future frames are excluded. Generated advice that cannot be delivered cleanly falls back to the original instruction; a policy/RNG/socket invariant failure is instead an infrastructure outcome. Simulation waits do not increase the action horizon and are not compute-matched to C/R/G.

Comparisons and uncertainty. For each policy and comparator, complete valid paired outcomes give four cells: both success, treatment-only success W, reference-only success L, and both failure. B:V–L is the visual-information contrast; B:R/G/L/V–C and base:V–C provide the prespecified contextual comparisons. The original base shares architecture and pretraining with B, so agreement across these two endpoints is within-lineage replication only. This earlier study did not run formal GR00T C/V evaluation; its separate two-case engineering tests are not transfer-performance evidence. The later bounded GR00T experiment is reported separately in Appendix M, with incomplete coverage retained.

Matched pairs require compatible initial fingerprints, complete exact pretrigger traces and matched trigger observations where reached. A valid early termination before the trigger is retained if its trajectory meets the paired contract; it does not require an assistant receipt. Actual L/V application requires linked CLI, image/modality, response and policy-instruction evidence. Delivery, clean fallback, early termination and technical failure are counted separately rather than conditioning the primary effect on successful delivery.

The full reduced-case denominator is 50 in each contrast. With complete matched evidence, net difference is $(W - L)/50$. Otherwise the report gives identification bounds by allowing each unresolved binary endpoint its feasible value; a usable one-sided endpoint tightens its case bound. Initial/prefix deviations cannot borrow unmatched treatment outcomes as though observed on the canonical C trajectory. Bounds are not confidence intervals. Fixed-seed task-block bootstrap intervals describe only available valid paired effects; they do not resolve missingness, assume deployment representativeness, or establish statistical superiority across six comparisons. One seed per task cannot quantify seed-to-seed variability within a task.

Final accounting. All 350 amended assignments have execution claims; 350 have individually valid terminal outcomes. B:V–L has 42/50 matched pairs, with 2 V-only and 0 L-only successes. Its full-cohort net-effect bounds are [-6.0, +16.0] percentage points; 8 comparisons remain unresolved. The following tables are populated from the final machine-readable report after terminal reconciliation; no pending value is imputed. Claimed assignments, valid outcomes, matched pairs and

assistant applications are distinct quantities. All raw evidence and earlier batch reports are retained; a terminated batch alone is not completion of the original 2,500-case plan. CLI usage is deduplicated by actual invocation across batches and includes retained out-of-cohort work. Reasoning tokens are part of output tokens, and missing counters remain unknown. The final paper reports negative and unknown outcomes without converting the follow-up into a revised historical or leaderboard score.

Three-batch execution and broker recovery. The retained batches contain 198, 99 and 244 claimed assignments, respectively. The first contains seven target arms and 191 out-of-cohort arms; the later 343 claims are disjoint remaining target assignments. This accounts for all 350 amended arms exactly once. The original dispatcher was administratively drained; the first continuation then stopped on a broker-transport recovery classification error. A separately frozen final infrastructure version accepts only verified completed transport-recovery receipts. It does not repeat a claimed episode or re-invoke an already executed CLI request. Earlier source versions, stop/error records and outcomes remain archived. This broker-transport recovery does not authorize a policy-socket reset or an extra simulation step.

Why complete arm outcomes do not imply complete paired effects. Every recorded arm has a valid native-reset terminal outcome, and no reduced-cohort arm is technically unknown. Nevertheless, each comparison has 3–8 initial-state deviations. These are valid episodes on differing realized scenes, not comparable outcomes on a common initial state. They remain in the fixed 50-case denominator and in the bounds, rather than being discarded or counted as treatment failures. The 191 outside-cohort outcomes are a different issue: they were excluded by the deterministic scope amendment and are retained in the archive, without entering the amended estimand. Neither record class is erased.

Disaggregating the two B:V–L wins. The full paired ledger identifies exactly two treatment-only successes in this contrast. In PickPlaceCounterToStove (fresh-24-00), B:C succeeds at 574/600 steps. B:L receives applied advice and fails at 600. B:V reaches its bounded wait, takes the declared unchanged-instruction fallback, and succeeds at the same 574 steps as C. This is a system-level V-arm win over L, not evidence that visual advice improved the controller. In WashFruitColander (fresh-47-00), B:V receives verified applied advice and succeeds at 2,347/3,150 steps; both C and L fail at the horizon. This is also the sole matched B:V–C treatment-only success. Neither example licenses an aggregate visual-grounding advantage: eight V–L pairs remain unresolved and its identification bounds include negative effects.

The original-base regression occurs in StoreLeftoversInBowl (fresh-41-00): C succeeds at 1,373 steps while V receives applied advice and fails at the 2,550-step horizon. This observed matched harm stays in the primary table. B:V has 20 valid early terminations, 20 valid advice applications and ten valid fallbacks; B:L has 19, 22 and nine, respectively. These totals concern all assigned arms; matched-pair subsets differ. In the 42 B:V–L matched pairs, V has 16 early terminations, 18 deliveries and eight fallbacks, versus 16, 19 and seven in L. The paired system contrast intentionally includes these fallbacks, rather than conditioning on delivery after seeing outcomes.

Complete comparison accounting. The pair categories and four outcome cells are shown separately; initial/prefix deviations are never silently recoded as failures. Bounds are percentage points on the fixed 50-case denominator.

Contrast	Matched	Unknown	Initial	Prefix	Both +	Treat. only	Ref. only	Both −
B: R–C	47	0	3	0	25	0	1	21
B: G–C	44	0	6	0	22	1	1	20
B: L–C	45	0	5	0	23	0	1	21
B: V–C	44	0	6	0	24	1	0	19
B: V–L	42	0	8	0	21	2	0	19
Base: V–C	47	0	3	0	23	0	1	23

Arm accounting and assistance. Every arm retains 50 assignments. Success and failure counts use individually verified endpoints; paired comparability is assessed in the previous table. Delivery and clean fallback are separate from task outcome.

Arm	Present	Success	Failure	Unknown	Early terminal	Valid delivery	Valid fallback
B:C	50	29	21	0	22	0	0
B:R	50	28	22	0	21	29	0
B:G	50	27	23	0	19	31	0
B:L	50	27	23	0	19	22	9
B:V	50	29	21	0	20	20	10
Base:C	50	26	24	0	22	0	0
Base:V	50	25	25	0	23	16	11

Conditional task-block sensitivity. These intervals use only valid matched pairs and exchangeable task-block resampling. They are not identification bounds, do not correct outcome missingness, and are not multiplicity-adjusted superiority tests.

Contrast	Tasks	Pairs	Conditional net (pp)	95% percentile range (pp)	Draws
B: R–C	47	47	-2.1	[-6.4, +0.0]	2000
B: G–C	44	44	+0.0	[-6.8, +6.8]	2000
B: L–C	45	45	-2.2	[-6.7, +0.0]	2000
B: V–C	44	44	+2.3	[+0.0, +6.8]	2000
B: V–L	42	42	+4.8	[+0.0, +11.9]	2000
Base: V–C	47	47	-2.1	[-6.4, +0.0]	2000

Actual resource accounting. The combined execution ledger contains 72 unique CLI invocations, including retained out-of-cohort work. Known input tokens are 1,244,218; known output tokens are 24,566, including 8,705 reported reasoning tokens. Input/output accounting is complete. Reasoning tokens are not added again. Recorded invocation latency sums to 871.88 seconds; this is not per-case latency or robot waiting time. No monetary cost is inferred.

J Historical failure-replay method and outcomes

J.1 A single scheduled language intervention

The policy replans every 16 environment steps. For horizon H, the assistance trigger is

$$\tau = 16 \lceil H/32 \rceil . \tag{S13}$$

This is the first regular query at or after half the horizon. The trigger depends only on the fixed action budget, not on hidden success state, retrospective failure information, or a learned stall score. The original failure list selects the *study cohort*; its outcome labels are not supplied to the assistant.

At τ, the wrapper sends the original task instruction, current left/right/wrist RGB images (256×256 each), and remaining environment steps to `gpt-6-astra` at `high` reasoning effort. Routing identifiers associate each response with its case. Hidden simulator state, success predicates, future frames, and original outcome labels are excluded. The robot policy continues receiving its ordinary proprioceptive observations; these are not additionally sent to Astra.

The supplement reproduces the fixed prompt, response-schema factory, and append rule. The response provides at most 1,500 characters of subgoal text. The wrapper preserves the original instruction, appends "Immediate next actions:" and that text, then appends "Then complete the original task." The augmented instruction persists for subsequent policy queries. Astra receives no second observation and produces no motor actions. Its generated rationale is not a verified explanation of failure.

The broker groups at most eight distinct requests into labeled image contact sheets. Each response must pass the recorded schema and identity checks. The prompt forbids tool use, and event traces are checked for tool calls. The CLI invocation records the requested model and reasoning effort; it is not an independently attested identity of service-side model weights. Batch context is a shared resource and a possible source of dependence between cases, not eight separate model invocations.

J.2 Budget and runtime invariants

Both arms preserve seed, task, native reset, endpoint, observation history, five Euler integration steps, replanning interval, crop ratio, and registered horizon. Simulation pauses while awaiting assistance, for at most 1,200 wall-clock seconds. This wait consumes additional inference and wall time but does not add action steps. The control is therefore *action-budget matched*, not compute- or latency-matched. Read-only RNG keepalives keep the same policy socket alive without reconnecting or resetting it. Missing, invalid, or unapplied responses remain technical outcomes; completed failures are not rerun to select a better outcome.

K Paired Replay Design

K.1 Historical record and fixed cohort

The historical native-reset evaluation contains 50 tasks with 50 episodes each: 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen, in pretraining scenes. Environment and policy seeds are fixed in $[2090900000, 2090902499]$; task horizons follow RoboCasa 1.0.1. All 2,500 planned episodes have terminal records, with zero missing or infrastructure-unknown outcomes. Every failure reaches its original horizon. The original Counter reset implementation is checked before and after each episode.

We define the rescue cohort before replay as

$$\mathcal{F} = \{i : Y_i^{\text{historical}} = 0\}, \qquad |\mathcal{F}| = 1{,}004. \tag{S14}$$

It spans 49 tasks; CloseFridge contributes no cases because its historical 50 episodes all succeeded. Each case is assigned two arms, giving 2,008 planned arm attempts but only 1,004 unique cases. The first campaign permanently claims 36 cases; an infrastructure continuation claims the other 968. This partition is fixed at whole-case level before continuation. No later outcome is substituted for a first-partition outcome.

Four infrastructure-development cases, indices 48, 97, 249, and 345, were run separately before the main study and yielded zero pilot rescues. Their pilot outcomes are excluded, but their identities remain in the primary cohort and are explicitly development-exposed. We report a leave-these-identities-out sensitivity analysis rather than describe the full cohort as untouched test data.

K.2 Verification gates and terminal categories

For each case, the verifier checks three distinct conditions. **Initial agreement** requires both replay initial fingerprints to agree with the historical case, including the recorded scene, observation, instruction, and state evidence. **Prefix agreement** requires exact paired state/image/instruction/action-query hashes before the intervention. **Applied assistance** requires linked request, image, response, CLI, and policy-query evidence showing that a valid Astra response actually reached the assisted arm. The state checks are offline measurements; their privileged contents are never policy or assistant inputs.

Let M_i indicate valid completed arms with matching historical initial fingerprints and paired preintervention evidence. Let A_i indicate verified applied assistance and $E_i = M_i A_i$. The rescue indicator is

$$R_i = E_i \mathbf{1}[Y_i^C = 0,\ Y_i^A = 1]. \tag{S15}$$

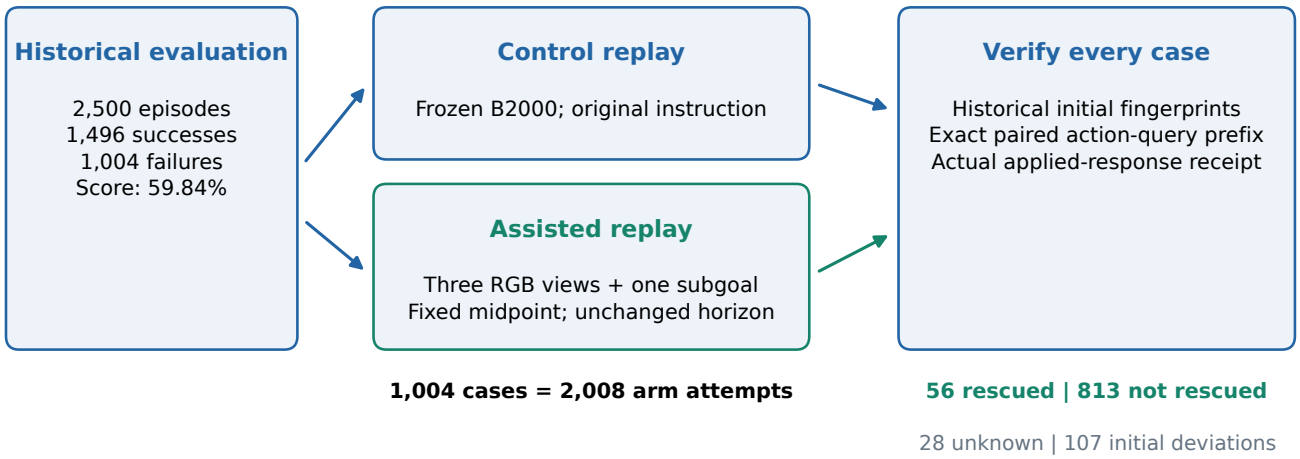


Figure S2. **Two records, one fixed failure cohort.** The original evaluation remains 1,496/2,500. Each of its 1,004 failures is assigned one control and one assisted replay. Assistance changes only the instruction at a fixed query; the robot weights and action horizon remain frozen. Offline replay checks and actual-response evidence determine whether the pair is eligible. All four terminal categories stay in the cohort denominator.

Here C and A denote the control and assisted outcomes. A valid response with no outcome, or an assisted success without matching evidence, is not a strict rescue. An eligible pair that does not satisfy the outcome transition is a non-rescue. Confirmed initial mismatches receive their own deviation category; otherwise missing or invalid evidence remains unknown. These categories are exhaustive and mutually exclusive in the settled ledger.

A further trace audit checks equal state and RGB hashes at the intervention query itself, before the changed instruction is evaluated. The checks strengthen a within-case comparison, but do not prove simulator equivalence outside the recorded contracts. In particular, preintervention traces are compared between the two new arms; they are not reconstructed historical trajectories. The historical archive contains outcome records and initial fingerprints, not videos of the original failures. We cannot retrospectively identify the original first physical error from a hash.

K.3 Coverage and recovery are different quantities

The primary estimand is the realized confirmed rescue fraction of the fixed historical cohort, $\widehat{r}_{\mathcal{F}} = \sum_i R_i/N$, with $N = 1004$. Its exact accounting identity is

$$\underbrace{\frac{R}{N}}_{\text{confirmed yield}} = \underbrace{\frac{E}{N}}_{\text{valid coverage}} \underbrace{\frac{R}{E}}_{\text{conditional recovery}}, \qquad E > 0. \tag{S16}$$

This decomposition distinguishes a missing valid comparison from an observed failure after valid assistance. Unknowns and deviations remain in N. The matched-complete count M is an intermediate evidence quantity; R/M is reported in the supplement. Neither R/E nor R/N is an overall deployment success rate. The 1,496 historical successes were not retested, so assistance-induced regressions on them are unmeasured.

The counts form a census of a failure-selected cohort. Task, timing, and batch analyses are retrospective, not preregistered tests. We report technical attrition directly, without assuming missing-at-random outcomes or imputing success. Zero control successes are an observed property of the eligible replays, not an eligibility requirement. The paired asymmetry does not establish full-population improvement.

L Results

L.1 Baseline performance and the opportunity set

Table 2 records the frozen endpoint. The 49.25-point Atomic-Seen versus Composite-Unseen gap motivates studying failures, but does not diagnose their causes. Composite-Unseen supplies 530 of the 1,004 failures (52.79%). GatherTableware, HeatKebabSandwich, and PanTransfer each fail all 50 historical episodes. These failures identify an opportunity set for investigation; they do not establish that an instruction-level correction is sufficient. The public numerical context in Table 3 does not enter this paired replay analysis.

Terminal category	Cases	Fraction of cohort
Strict rescue	56	5.58%
Eligible non-rescue	813	80.98%
Technical unknown	28	2.79%
Initial-state deviation	107	10.66%
Fixed cohort	1004	100.00%

Table S3. **Complete accounting, incomplete valid coverage.** All planned attempts are terminal. Unknowns and deviations are not converted to policy failures or removed from the denominator; rounded percentages need not sum exactly.

L.2 RQ1: How much of the cohort is verifiable?

Table S3 gives the complete ledger. We obtain $M = 874$ matched completed replay pairs and $E = 869$ pairs with verified applied assistance. Thus $86.55\% \times 6.44\% = 5.58\%$ up to rounding: the conditional result alone omits the 13.45% without a valid intervention comparison. Completed accounting does not mean complete valid coverage.

Among eligible pairs, the complete paired table is 56 assisted-only successes and 813 failures in both arms, with zero control-only and zero both-arm successes. The 874-pair raw matched table adds five both-failure pairs whose assistance evidence is invalid; those five remain unknown, not eligible non-rescues. Their presence illustrates why terminal success bits alone cannot establish that an intervention was tested.

The first partition yields 0 rescues, 3 non-rescues, 28 unknowns, and 5 initial deviations. The continuation yields 56, 810, 0, and 102 respectively. All 28 unknowns therefore arise in the first partition. The continuation changed infrastructure handling, including same-socket keepalives and continued dispatch after validated native-reset deviations. We retain partition membership because this operational history can affect technical coverage. The study is not an uninterrupted homogeneous execution, although its endpoint, action budget, and intervention rule are fixed.

L.3 RQ2: Does advice change actions and outcomes?

At the intervention query, all 869 eligible pairs have identical recorded state and RGB hashes, different instruction hashes, and immediately different returned action hashes. The action hash covers the returned array's shape, dtype, and bytes; it does not include the instruction. Thus zero-step action-output divergence is observed in all 56 rescued and all 813 non-rescued pairs, with no missing trigger record. This establishes a computational difference, not its magnitude or usefulness. Hashes cannot distinguish a meaningful

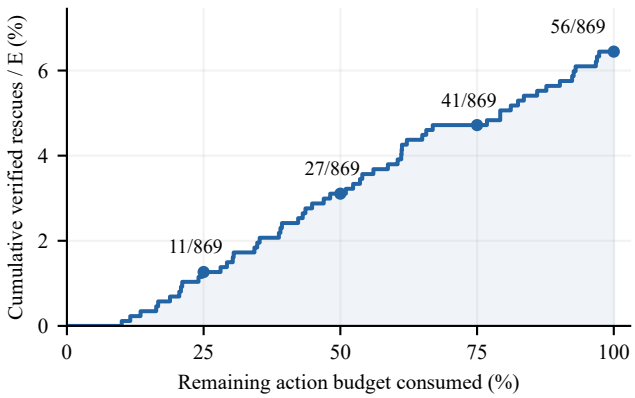


Figure S3. **Recovery takes continued action.** At 25/50/75/100% of the remaining budget, 11/27/41/56 cases have succeeded, out of 869 eligible pairs. All 813 eligible non-rescues reach their original horizons. The curve describes recorded success timing, not performance under alternative intervention times.

motor correction from a tiny numerical perturbation, and this comparison does not independently randomize service scheduling or arithmetic effects.

To measure recovery timing, define $u_i = (T_i - \tau_i)/(H_i - \tau_i)$ for a rescued pair, where T_i is its recorded success step. Figure S3 plots

$$G_E(u) = \frac{1}{869} \sum_{i:R_i=1} \mathbf{1}[u_i \le u], \tag{S17}$$

with zero contribution from non-rescues. Unlike a CDF conditioned only on success, this curve retains all eligible failures in its denominator and ends at 6.44%, not 100%.

Among the 56 rescues, median u_i is 51.67% (interquartile range 30.45–77.41%), and the median continuation is 361 environment steps. Fifteen rescues finish in the last quarter of their remaining budget; seven use more than 90%. A short observation window after advice would therefore miss recorded successes. An earlier trigger might help, harm, or leave outcomes unchanged; that counterfactual has not been tested.

L.4 RQ3: Where does recovery remain limited?

Table S4 and Fig. S4 expose two different limitations. Valid coverage is 89.54%, 75.70%, and 92.26% in Atomic-Seen, Composite-Seen, and Composite-Unseen; conditional recovery is 8.76%, 10.29%, and 3.89%. The lowest yield in Composite-Unseen is therefore not the arithmetic consequence of lower coverage. Conversely, Composite-Seen loses 70/321 cases to initial-state deviations, compared with 21/530 in Composite-Unseen. These patterns separate the accounting factors without identifying why the splits differ.

Original failure split	Fixed cases	Eligible E	Rescued	Unknown	Initial deviation	Rescued/fixed cases
Atomic-Seen	153	137	12	0	16	7.84%
Composite-Seen	321	243	25	8	70	7.79%
Composite-Unseen	530	489	19	20	21	3.58%
All	1004	869	56	28	107	5.58%

Table S4. **Task familiarity and technical coverage.** Counts are conditioned on historical failure. The lower confirmed rescue fraction in Composite-Unseen is descriptive: the splits have different tasks and horizons, and no controlled generalization effect is identified. In particular, Composite-Seen has a much higher initial-deviation count.

Rescues span 27 of the 49 tasks with historical failures. BreadSelection and PrepareCoffee account for 13/56 rescues (23.21%); the five largest rescue counts contribute 24/56 (42.86%). Equal weighting of failure-containing tasks gives 7.83%, compared with the pooled 5.58%; these weight different task mixtures. Complete rows are retained in the supplement.

The three originally zero-success tasks yield one rescue in 150 failures (135 eligible pairs); the five lowest historical task scores yield one in 237 (219 eligible). These difficult subsets remain unresolved. However, the full task scatter does not support a monotonic difficulty law: historical task success has descriptive Spearman correlation 0.075 with fixed-cohort yield (49 tasks), and 0.117 with eligible-pair recovery (48 tasks). The bottom-five finding is a retrospective subset description, not a general predictor of recoverability.

Whole-task resampling within original splits gives a 3.59–7.95% percentile range for confirmed yield. This exploratory task-mixture sensitivity is not a confidence interval for the finite census or a causal effect; its assumptions and the separate batch sensitivity are detailed in the supplement.

Development exposure. Removing all four disclosed pilot identities yields 55/1,000 confirmed rescues (5.50%); this sensitivity does not replace the primary cohort. The separate pilot's 0/4 outcomes are never added to it.

L.5 Two matched outcomes within the same task

Figure S5 shows the complete two-case historical OpenDrawer failure subset. In case 186 (seed 2090900186), Astra supplies a subgoal about approaching the visible drawer handle, grasping it, and pulling outward. The applied response is linked to the policy's changed instruction. Its assisted arm succeeds at step 674, while the unassisted control exhausts the 750-step horizon. Case 586 (seed 2090900586) also receives valid advice but neither arm succeeds. Equal initial and paired-prefix evidence localizes the experimental difference to the intervention period under the checked contracts. It does not tell us which phrase, visual interpretation, or subsequent physical motion produced the positive outcome; the recorded trigger images are not a postintervention trajectory.

Primary-study accounting	Recorded value
Unique CLI invocations	135 (134 successful)
Known input tokens	2,592,713
Known output tokens	195,237
Reported reasoning tokens	75,511
Median CLI time / invocation	46.16 s
Median eligible-case request wait	191.38 s
95th percentile eligible-case wait	287.39 s

Table S5. **Actual inference overhead.** Invocations are deduplicated across batches and receipts. One error lacks counters; complete token totals and dollar cost are unknown. Reasoning tokens are not added again to output tokens. Request waiting includes queue/transport time and is distinct from CLI runtime.

L.6 Operational cost of the intervention

Table S5 distinguishes request waiting from CLI runtime. Total known invocation latency is 6,186.44 s; batching prevents interpreting this as per-case latency. Eligible-case waits are available for 869 cases, excluding timed-out and invalid cases. The successful OpenDrawer case waits 327.17 s while its batch executes for 48.98 s. The separate pilot adds 19,071 input and 629 output tokens (137 reasoning tokens) and 24.35 s of CLI runtime. Failed and unapplied responses remain in accounting. Verified monetary cost is unavailable.

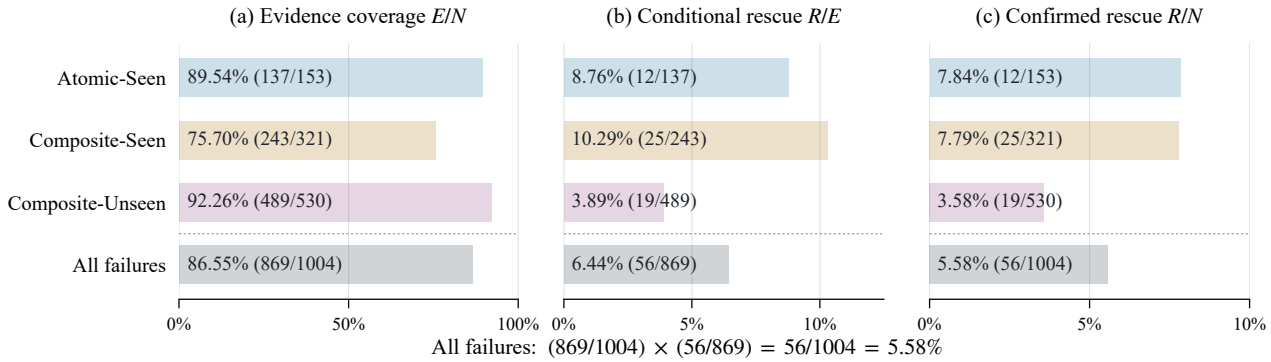


Figure S4. **Technical coverage and behavioral recovery separate.** Each row factorizes the confirmed yield under Eq. S16. Composite-Unseen has the highest eligible coverage and the lowest conditional rescue fraction. This is a descriptive split comparison, not a controlled effect of task novelty.

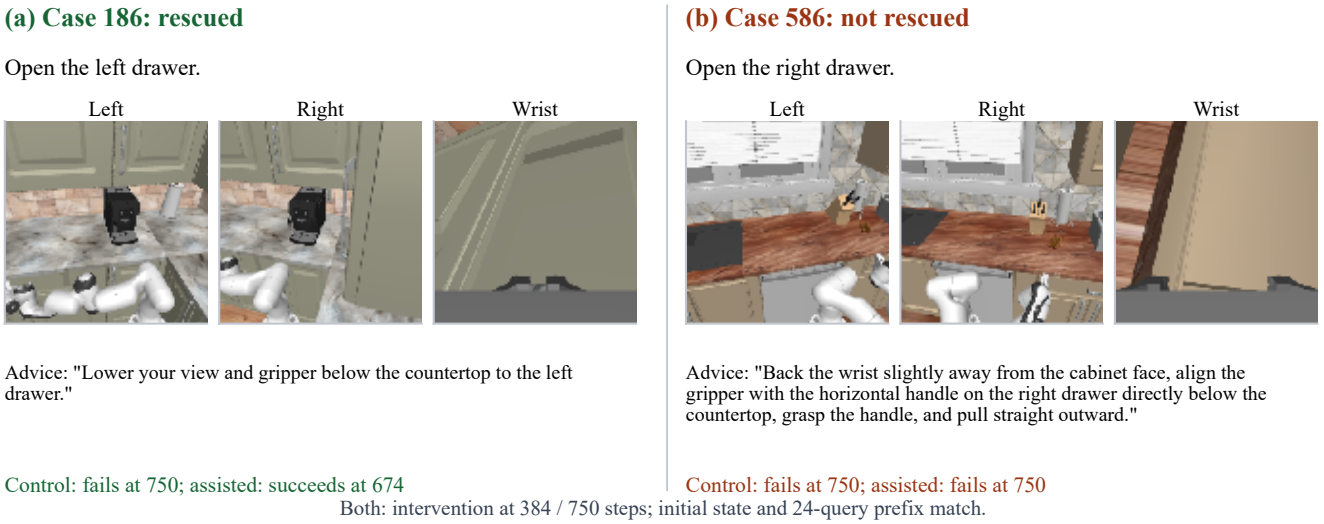


Figure S5. **Actual intervention observations for both historical OpenDrawer failures.** The rows show the original three RGB views at the trigger, not reconstructed videos or generated scenes. Both pairs match their historical initial fingerprints and have 24 exact paired preintervention queries; both trigger at step 384 with horizon 750. Case 186 is rescued (assisted success at 674; control fails at 750). Case 586 is not (both fail at 750). These observations and instructions illustrate two outcomes of the same assistance interface; they do not establish a physical causal explanation.

M Bounded Exploratory Endpoint and Intervention Diagnostic

This additional diagnostic allocated one seed to each of the 50 target tasks. Its six planned assignments per case are the original Xiaomi checkpoint (X), A1613 Action LoRA (A), frozen B2000 (B), B with a visual teacher instruction (BV), and a GR00T candidate without/with that intervention (GRC/GRV). Thus the target was 300 arm assignments over 50 unique task–seed cases, not 300 independent cases. Final execution produced 262 valid terminal outcomes; 38 assignments remained unstarted. This bounded exploratory study is neither a new official benchmark nor a multi-seed confirmation of an adaptation benefit.

Selection and execution. The fixed cohort contains all 50 target tasks, with one fresh seed per task (2027101100 through 2027101149), native reset and unchanged task horizons. X, A and B use frozen endpoints. The planned six-arm cohort includes B with one visual-teacher subgoal and GR00T without/with the same bounded intervention. Evaluation stopped under the preset time budget; unstarted assignments are retained separately.

Arm	Success	Failure	Unknown	Invalid	Not run
X	30	20	0	0	0
A	32	18	0	0	0
B	33	17	0	0	0
BV	32	18	0	0	0
GRC	6	25	0	0	19
GRV	7	24	0	0	19

Table S6. Final assignment accounting in the bounded diagnostic. Every row has the fixed denominator 50. Unknown and invalid outcomes are not recoded as behavioral failure.

Terminal status and initial-state compatibility are different checks. Failure follows the frozen evaluator: a full-horizon failure or a verified natural environment termination/truncation; an administrative time limit is unresolved. The X/A/B comparisons use known success/failure outcomes and equal verified initial-state fingerprints. Within-policy BV–B and GRV–GRC comparisons additionally require the strict audit of equal pre-intervention query traces and trigger observations, or equal early termination before intervention. All other comparisons remain unresolved. Different policies are expected to diverge after the common reset; action-prefix equality is not imposed across X/A/B.

Pair	Win	Loss	Tie	Unres.	Net bound (pp)
A–X	5	2	38	5	[-4.0, +16.0]
B–A	5	4	35	6	[-10.0, +14.0]
B–X	5	2	37	6	[-6.0, +18.0]
BV–B	1	1	42	6	[-10.0, +2.0]
GRV–GRC	1	0	28	21	[-36.0, +44.0]

Table S7. First-named endpoint relative to the second, over the fixed 50-case cohort. X/A/B bounds conservatively leave both outcomes unresolved for incompatible comparisons. Within-policy bounds follow the strict audit and retain any usable canonical control outcome. These are identification bounds, not confidence intervals.

One realization per task does not establish a reliable task-level success rate or a training-seed effect. This comparison does not isolate auxiliary stage supervision: that attribution requires a separately trained, matched stage-loss ablation. The bounded cohort also does not replace the earlier 600-pair comparison. Matching initial states supports a comparison of full deployment policies; it does not, by itself, verify identical intervention prefixes or isolate the causal contribution of a teacher response.

Cross-architecture admission. The admitted GR00T model identity is GR00T N1.5 RoboCasa multitask checkpoint-120000 (revision c484448), using its native 16-dimensional state representation and a 12-dimensional environment action. GR00T results are kept distinct from the Xiaomi-derived endpoints; different inherited models and interfaces prevent an architectural attribution from a score difference.

The deadline left 19 control and 19 intervention assignments unstarted. This is incomplete coverage of the fixed cohort, not a random subsample; completed-case success rates are not substituted for a complete score. The strict matched GR comparison is exploratory and cannot establish cross-architecture generalization.

Untrained intervention-selector prototype. An offline software prototype was prepared for estimating the incremental value of an intervention relative to continuing the frozen policy. It was not trained and was not evaluated in closed-loop robot execution. Its software checks do not establish intervention efficacy.

Fifteen synthetic software checks passed. No optimizer update, empirical calibration or closed-loop gate evaluation was performed; hand-specified demo probabilities are not learned predictions.

Offline initial-state diagnosis. The X/A/B initial states match in 42 of 50 cases. Inspection of the 24 saved snapshots for the remaining eight cases finds free-object joint translation differences in every case; the maximum across the three policy pairs within each case ranges from 0.968 to 1.917 m. These are differences of the saved free-joint translation coordinates, not full world-frame forward-kinematic distances. Seven cases have byte-identical XML. PreSoakPan additionally selects different pan assets (`pan_1` versus `pan_15`). WashLettuce has identical 14-dimensional proprioception despite a 0.992 m container-joint translation difference. Thus the discrepancies cannot be dismissed as texture or pixel differences. Layout, style and instruction are identical within all eight cases. The snapshots do not establish which random-number use or initialization path caused divergence; no fix or reclassification is claimed. Future comparisons should preregister checks of the complete physical state and asset hashes, and log reset-stage random states, asset choices, placement regions, object poses and robot bases to test deterministic initialization. A state-snapshot restoration study would require a separate, explicitly labeled protocol and would not replace the native-reset benchmark.

Applied-teacher outcomes. BV records 25 verified teacher applications, with one rescue and one regression among strictly matched comparisons; GRV records 28 verified teacher applications, with one rescue and no regressions among strictly matched comparisons. A matched rescue requires an actually delivered teacher response; fallback successes are not counted as teacher rescues. These small counts do not establish a general intervention benefit.

Study	Calls	Input	Output	Seconds
BV	25	430,501	7,548	306.53
GR	28	482,162	7,608	322.55

Table S8. Complete teacher-invocation inventory for this diagnostic. Seconds sum individual invocation latencies, not experiment wall time. Reasoning tokens are included in output counts and are not added again; dollar cost is unavailable.

Scope and unchanged historical results. The final receipt counts 53 new Astra invocations. No policy-training updates or real-robot trials were performed in this diagnostic. A possible future SO-101 experiment requires its own observation, action and embodiment adaptation; a Xiaomi or GR00T checkpoint cannot be assumed to control that robot directly. The historical author-run benchmark remains 1,496/2,500 (59.84%), and the historical failure-conditioned rescue result remains 56/1,004. These exploratory rows cannot be added to either denominator. The study provides no claim of a new rank, a trained gate, or a conference acceptance.

The final execution receipt verifies that the diagnostic's own evaluation and model processes had ended before delivery.